Shortcomings and Proposed Improvements in ComHAI and PLACO

A Critical Analysis of Human-AI Team Combination Methods

1. Humans Are Assumed Independent

1.1. The Problem

- The combination $\prod_i \phi^{[i]}_{h_i(x),j}$ treats every human as a fully independent source — assuming $P(h_i, h_j \mid y) = P(h_i \mid y) \cdot P(h_j \mid y)$
- Annotators from similar backgrounds share systematic biases and make correlated errors — agreement between them is weaker evidence than the formula assumes
- The method becomes **overconfident** when correlated humans agree, even if they are agreeing for the same wrong reason

1.2. The Fix

- Estimate **pairwise error correlations** from training data — how often do two humans make the same mistake on the same instance?
- Cluster humans by error similarity; treat near-identical annotators as **one voice** — select at most one per cluster
- Replace the independent product with a **covariance-adjusted likelihood** that discounts redundant agreement; uses only already-available training data

2. One Reliability Profile Per Human, Forever

2.1. The Problem

- A single global $\phi^{[i]}$ is estimated once and applied uniformly to every future instance — regardless of instance type
- Human accuracy is clearly **instance-dependent**: a radiologist brilliant at chest X-rays may be poor at brain scans; the model cannot know this
- The method over-trusts generalist humans on specialist instances and under-trusts specialists whose overall average is modest

2.2. The Fix

- **Lightweight:** cluster instances by difficulty or feature similarity; estimate one $\phi^{[i]}$ per cluster per human; at test time look up the cluster the current instance belongs to
- **Powerful:** embed instances and humans into a shared feature space and learn $\phi^{[i]}(x)$ directly — a small auxiliary network on top of the existing AI model, trained on the same labelled data
- Both versions require no new annotation and slot into the existing pipeline without changing the combination formula

3. Human Labels Guessed Once, Never Revised

3.1. The Problem

- PLACO estimates $h_i(x) = \arg\max_l \sum_y \phi^{[i]}_{ly} \cdot m^\theta_y(x)$ in a single pass — if $m^\theta(x)$ is slightly off, all human estimates are off, and the wrong humans get selected with no correction
- Estimation accuracy is only ∼30%; estimates per human are computed in complete isolation — no cross-human information sharing

3.2. The Fix

- Replace the one-shot guess with **iterative EM** (Dawid-Skene): alternate between updating the posterior over the true label and re-estimating each human's likely label until convergence
- Each round sharpens both estimates simultaneously — errors from the AI's initial output get corrected progressively rather than propagating
- Well-established in crowdsourcing literature; notably absent here despite both papers explicitly citing that work

4. Close-Call Pseudo-Label Commitment

4.1. The Problem

- A hard $\arg\max$ commits to one y^* before any human is queried; all selection is then oriented around that single answer
- When the top two classes are nearly equal (e.g. 52% vs 48%), the pseudo-label is a coin flip — and every downstream step inherits that error
- These ambiguous instances are **exactly** where human input matters most; the method is weakest precisely there

4.2. Fix A: Confidence-Gated Pairwise Selection

- If the top class exceeds a threshold (e.g. 65%) proceed as normal — zero change for the confident majority
- Below threshold: identify the top two candidate classes and select humans based on their **pairwise discrimination** ability for that specific pair, read directly from the existing confusion matrices — no new data needed

4.3. Fix B: Sequential Adaptive Querying

- On a close call, query one cheap human first; use their response to update the posterior — ambiguity often resolves after a single targeted query
- Only then select remaining humans using the now-sharper pseudo-label; fits naturally into the existing PLACO per-query cost model

5. Confusion Matrices Are Frozen in Time

5.1. The Problem — Three Failure Modes

- **Fatigue:** a human in hour four of labelling performs worse than at hour one — the frozen matrix reflects their long-run average, not current state
- **Skill gain:** a junior annotator improves significantly over months; the stale matrix underestimates them and the system chronically passes them over
- **Domain shift:** a hospital switches scanner models — every confusion matrix was built on the old images; the

system has no way to detect the shift

5.2. The Fix: Dual Rolling Window

- Maintain two matrices per human: a **long-run matrix** (all history) and a **recent-form matrix** (last N verified predictions)
- Blend them at inference time — weight shifts toward recent-form when the two diverge significantly, flagging fatigue, improvement, or domain shift
- Ground truth for the recent window accumulates naturally from verified predictions — **no extra annotation cost**; N is tunable per domain

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6. Hard Pseudo-Label Fails on Ambiguous Instances

6.1. The Problem

- A hard $\arg\max$ commits to one y^* before any human is queried — all human selection is then oriented around that single answer
- On easy instances the AI is confident and y^* is almost certainly correct — the method works well here
- On hard instances the AI outputs something like 52% wolf, 48% husky — y^* is essentially a coin flip, and every downstream step inherits that error with no recovery mechanism
- These ambiguous instances are **exactly** where human input matters most — the method is structurally weakest precisely there

6.2. Fix A: Confidence-Gated Pairwise Selection

- Compute the gap between the top two class probabilities after the AI forward pass
- If the top class exceeds a threshold (e.g. 65%) — commit to y^* and proceed as normal; zero change for the confident majority of instances
- Below threshold: identify the top two candidate classes and select humans based on their **pairwise discrimination** ability for that specific pair — how reliably does this human distinguish wolf from husky?
- This is a different lookup into the **existing confusion matrices** — no new data, no new architecture required

6.3. Fix B: Sequential Adaptive Querying

- On a close call, query one cheap human first rather than selecting a full subset upfront
- Use their response to update the AI posterior — ambiguity often resolves after a single targeted early query, saving the cost of a full subset
- If ambiguity persists, query one further targeted human and repeat; each query sharpens the pseudo-label before the next selection decision
- Fits naturally into the existing PLACO per-query cost model — no architectural changes needed

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